

Processando config 1, seed 2, teste: Alphabit  
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 2    |
| Value of the statistic       | : | 3.10 |
| p-value of test              | : | 0.21 |

-----  
CPU time used : 00:00:00.01

Generator state:  
16777216 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 17.17  
p-value of test : 0.03

-----  
CPU time used : 00:00:00.02

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 135.46  
p-value of test : 0.31

-----  
CPU time used : 00:00:00.01

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32498.30  
p-value of test : 0.85

-----  
CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 216.09  
p-value of test : 0.02

-----  
CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 301.13  
p-value of test : 0.55  
-----

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 1.82  
p-value of test : 0.03  
-----

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

Number of degrees of freedom : 32  
ChiSquare statistic : 16.44  
p-value of test : 0.99

-----  
Test on the values of the Statistic M

Number of degrees of freedom : 31  
ChiSquare statistic : 32.52  
p-value of test : 0.39

-----  
Test on the values of the Statistic J

Number of degrees of freedom : 32  
ChiSquare statistic : 25.57  
p-value of test : 0.78

-----  
Test on the values of the Statistic R

Number of degrees of freedom : 25  
ChiSquare statistic : 28.62  
p-value of test : 0.28

-----  
Test on the values of the Statistic C

Number of degrees of freedom : 15  
ChiSquare statistic : 11.91  
p-value of test : 0.69

-----  
CPU time used : 00:00:00.06

Generator state:  
134217728 bits have been read.

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HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config1\_seed2.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

Number of degrees of freedom : 62

ChiSquare statistic : 1771.67  
p-value of test : eps \*\*\*\*\*

-----  
Test on the values of the Statistic M

Number of degrees of freedom : 62  
ChiSquare statistic : 998.10  
p-value of test : eps \*\*\*\*\*

-----  
Test on the values of the Statistic J

Number of degrees of freedom : 160  
ChiSquare statistic : 315.81  
p-value of test : 2.8e-12 \*\*\*\*\*

-----  
Test on the values of the Statistic R

Number of degrees of freedom : 57  
ChiSquare statistic : 61.04  
p-value of test : 0.33

-----  
Test on the values of the Statistic C

Number of degrees of freedom : 32  
ChiSquare statistic : 29.29  
p-value of test : 0.60

-----  
CPU time used : 00:00:00.05

Generator state:  
150994688 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.20  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test                      | p-value |
|---------------------------|---------|
| 9 RandomWalk1 H (L = 320) | eps     |
| 9 RandomWalk1 M (L = 320) | eps     |
| 9 RandomWalk1 J (L = 320) | 2.8e-12 |

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All other tests were passed

